



Ithynx Solution Pvt Ltd.

AI - ML - DS - Cloud - Personality - Full Stack - LLM and many more

We Bring The Industry To Your Footstep...

WHAT YOU SEE

Presenting an industry-driven, hands-on training initiative meticulously designed to bridge the critical gap between traditional college curricula and rapidly evolving tech industry demands. Our focus centers extensively on high-growth, cutting-edge fields including Artificial Intelligence (AI), Advanced Full-Stack Development, Software Engineering, and Enterprise Architecture.

The standout pillar of our approach is a radical, project-based delivery model—built on a **70% hands-on lab and 30% concept split**—which eliminates passive listening and exposes students directly to real-world development environments. By replacing slide-heavy lectures with live coding, interactive hackathons, and production-grade capstone builds, we ensure every student moves from theory to execution.

KEY OBJECTIVES



Industry Relevance

Close the gap between what colleges teach and what the industry demands



Zero Faculty Burden

70% hands-on labs · 30% concept · 0% passive listening



Hands on Experience

Every student builds and deploys a real AI product before the program ends



Knowledge Filling

Making sure to fill in the gaps and doubts of candidates regarding work environments.

WHO ASSISTS YOU



200+
Students
Trained



3+ Years
Experienced
Coaches



Industry Aligned
Minds



Work Place
Exposed
Individuals

ACADEMIC PARTNERSHIP PROPOSAL · CONFIDENTIAL

Transform Your Students Into AI-Ready Engineers

A hands-on, industry-driven program in AI · ML · Deep Learning · Reinforcement Learning
Generative AI · Agentic AI · LLMs · Full-Stack AI Development

Presented exclusively to: **St. Joseph's College Of Engineering, Chennai**

By: iThynx Solutions Pvt Ltd | Your Growth, Our Mission

01 EXECUTIVE SUMMARY

Engineering colleges across India face a growing crisis — students graduate with strong fundamentals but zero hands-on exposure to the technologies every company is hiring for right now. AI, Machine Learning, Deep Learning, and Generative AI are not future skills. They are today's hiring baseline.

iThynx Solutions Pvt Ltd is a specialized technical training company that partners exclusively with engineering institutions to close this gap. We don't run tuition centres. We don't teach competitive exams. We do **one thing** — build industry-ready AI engineers inside your campus.



Our Mission

Close the gap between what colleges teach and what the industry demands



Our Method

70% hands-on labs · 30% concept · 0% passive listening



Our Promise

Every student builds and deploys a real AI product before the program ends

02 WHY THIS MATTERS RIGHT NOW

The AI job market has fundamentally shifted. These are not projections — this is today's reality:

97M+

AI-related jobs globally by 2025 (World Economic Forum)

72%

Employers say new grads lack practical AI skills (LinkedIn)

4.2×

Salary premium for AI-skilled engineers vs. peers

#1

Most in-demand skill in India's tech sector — 3 years running

Your students are competing against peers from IITs and NITs who already have this exposure. This program levels the playing field — and in many cases, puts your students ahead.

03 THE PROBLEM WE SOLVE FOR YOUR INSTITUTION

✗ CURRENT REALITY IN MOST COLLEGES	✓ WHAT iThynx DELIVERS
Slide-heavy lectures with no code	Live coding every session — students code along in real time
Syllabus 3–5 years behind industry	2026 curriculum — updated every quarter with industry input
No GenAI, LLM, or Agentic AI exposure	Dedicated full modules on ChatGPT stack, RAG, Agents
Assessment = end-of-semester written exam	Project-based assessment + GitHub portfolio as proof-of-work
Faculty stretched thin, no AI expertise	We bring our own trainers — zero burden on your faculty
No placement differentiation for AI roles	Students leave with deployed projects recruiters can actually see
No NBA / NAAC documented industry tie-up	Formal MOU + activity reports for accreditation documentation

04 DIRECT BENEFITS TO YOUR INSTITUTION



For Management & Leadership

- ◆ Formal MOU + documented industry collaboration for NBA / NAAC files
- ◆ Institutional branding on all certifications and event materials
- ◆ No infrastructure cost — we bring trainers



For Students & Faculty

- ◆ Students build real, deployed AI projects — shareable with recruiters
- ◆ GitHub portfolio that speaks louder than any certificate
- ◆ Monthly industry expert sessions, hackathons, and paper opportunities

05 HOW WE WORK — OUR DELIVERY MODEL

We integrate into your academic schedule without disrupting it. Our trainers come to your campus (or deliver online), manage everything end-to-end, and produce documented outcomes. Here's exactly how every session works:



Live Coding

Instructor codes live — students replicate in real time on Colab/Jupyter. No death-by-slides.



Lab Sprint

Weekly guided lab challenge on a real dataset. Auto-graded. Immediate feedback.



Hackathon

Bi-cohort team hackathon simulating real industry sprints. Best projects win awards.



Capstone Build

Final 4-week intensive — students ship a real AI product and present to a panel.

06 ENGAGEMENT MODELS — CHOOSE WHAT FITS

FORMAT	BEST SUITED FOR	WHAT'S INCLUDED
 AI Sprint Workshop	Tech Fests · Induction Week · One-day event	Intensive intro session, demo, certificates — high-energy, zero prep needed from you
 Semester Add-on Module	3rd / 4th year elective or value-add batch	Full curriculum delivery, labs, assessments, certificates — runs parallel to academics
 AI Immersion Program	CSE / IT / ECE departments seeking deep skill	End-to-end program: all 7 tracks, capstone, demo day, placement prep, GitHub portfolio
 Centre of Excellence	Institutions wanting long-term AI brand identity	AI lab setup, faculty training, annual curriculum, branding, research paper co-authorship

07 WHY iThynx — NOT ANOTHER TRAINING VENDOR

TYPICAL TRAINING VENDOR

iThynx EDTECH

Generic content, reused for all colleges	Proposal and curriculum customized for your department & student profile
Trainers are moonlighting professionals	Dedicated trainers with active industry roles — not retired instructors
Certification with no market recognition	Portfolio of deployed projects + GitHub — what recruiters actually look for
No documentation for college records	Full outcome reports, attendance logs, MOU — ready for accreditation
You manage logistics	We handle everything — scheduling, labs, grading, demo day, awards

MODULE 01



MACHINE LEARNING

From Raw Data to Intelligent Predictions — The Foundation of All AI

OVERVIEW

Machine Learning is the backbone of every AI application — from Netflix recommendations to fraud detection. Students learn to take messy, real-world data and turn it into working predictive systems. No theory-only lectures. Every concept is immediately coded and tested on real datasets from domains students recognise — healthcare, finance, e-commerce, social media.

85%

Of all AI roles require ML proficiency

3×

Higher interview shortlisting for ML portfolio holders

₹8–18 LPA

Typical starting range for ML engineers in India

Week 1

Students write their first predictive model

CURRICULUM

CONCEPT & SKILL	HANDS-ON PROJECT / LAB
Python for AI — NumPy, Pandas, Matplotlib	EDA dashboard on a real Kaggle dataset — insights presented to class
Supervised Learning — Regression	House price predictor: trained, evaluated, deployed as web app
Supervised Learning — Classification	Spam email classifier with 95%+ accuracy on live email data
Unsupervised Learning — Clustering & PCA	Customer segmentation engine — visual cluster report
Feature Engineering & Data Preprocessing	Clean and transform a deliberately messy industry dataset
Model Evaluation — Accuracy, F1, ROC-AUC	Comparative model scorecard: 5 algorithms head-to-head
Cross Validation & Hyperparameter Tuning	GridSearchCV pipeline — students find the best model config
ML Pipelines & API Deployment	Deploy trained model as a REST API — accessible via browser URL

TOOLS & TECH



Python 3.12



Scikit-learn



Pandas / NumPy



FastAPI + Streamlit



WHAT STUDENTS WALK AWAY WITH

- ◆ 3 fully deployed ML models with live URLs — shareable on LinkedIn and resume
- ◆ Complete GitHub repository with clean notebooks, README, and documented code
- ◆ Ability to explain and defend model choices in a technical interview
- ◆ iThynx Machine Learning Certification — industry-acknowledged

MODULE 02



DEEP LEARNING

Neural Networks at Scale — How AI Learns to See, Hear, and Understand

OVERVIEW

Deep Learning is what powers face recognition, voice assistants, self-driving cars, and medical imaging AI. Students don't just learn what a neural network is — they build one from scratch using only NumPy, then progressively scale to CNNs, RNNs, and Transformer architectures. The moment students watch their network learn in real time — they're hooked.

92%

Of vision & NLP tasks use
DL models

₹12–25 LPA

DL engineer compensation
range in IndiaFrom
ScratchStudents build a neural net
with zero frameworks first

Deploy Day

Every student has a live
model URL by end of
program

CURRICULUM

CONCEPT & SKILL	HANDS-ON PROJECT / LAB
Neural Networks from Scratch	Build digit recognizer (MNIST) using only NumPy — no PyTorch
Backpropagation & Optimizers	Visualize gradient descent live — see loss curves in real time
Convolutional Neural Networks (CNN)	Real-time cat vs dog classifier — webcam integration demo
Transfer Learning & Fine-Tuning	Fine-tune ResNet50 on a custom student-chosen image dataset
Recurrent Networks & LSTMs	Sentiment analysis on Amazon/Flipkart product reviews dataset
Attention Mechanism & Transformers	Code a mini-transformer — understand BERT from the inside out
Regularization — Dropout, BatchNorm, Augmentation	Prevent overfitting — before/after comparison shown live

Model Deployment on Cloud

Host DL model on Hugging Face Spaces — share public link

TOOLS & TECH



PyTorch



Hugging Face



Google Colab (GPU)



OpenCV



WHAT STUDENTS WALK AWAY WITH

- ◆ A computer vision app live on the internet — built entirely by the student
- ◆ Deep understanding of how transformer models work internally — rare and valued
- ◆ iThynx Deep Learning Certification
- ◆ Capstone: Fine-tuned model on domain-specific data — submitted as final project

MODULE 03



REINFORCEMENT LEARNING

Teaching Machines to Make Decisions — Through Trial, Error, and Reward

OVERVIEW

Reinforcement Learning is how AlphaGo beat the world chess champion, how robots learn to walk, and how recommendation engines optimise for engagement. Students train agents that literally learn by doing — and watching an agent go from random actions to winning behaviour in real time is the most memorable moment in the entire program. This module builds intuition for decision-making under uncertainty that applies far beyond AI.



AlphaGo, ChatGPT RLHF,
Tesla Autopilot — all RL

Live Demo

Students watch their agent
learn in the same session

Robotics & Gaming

Two fastest-growing RL
application domains

Rare Skill

<5% of graduates have any
RL exposure

CURRICULUM

CONCEPT & SKILL	HANDS-ON PROJECT / LAB
RL Fundamentals — Agent, Env, State, Reward, Policy	Design and solve a custom GridWorld environment from scratch
Markov Decision Processes (MDPs)	Solve a maze using value iteration — visualize the value function
Q-Learning & SARSA	CartPole balancing agent — see it fail, then learn, in real time
Deep Q-Networks (DQN)	Train a neural network to play Atari Pong — live game footage
Policy Gradient Methods (REINFORCE)	LunarLander agent — from crashing to landing in 30 minutes
Actor-Critic Methods & PPO	Continuous control — robot arm simulation on-screen
Multi-Agent RL	Competitive and cooperative agent scenarios — simulation shown live

TOOLS & TECH



OpenAI Gymnasium



Stable-Baselines3



PyTorch



TensorBoard



WHAT STUDENTS WALK AWAY WITH

- ◆ 4 trained RL agents on progressively harder environments — all logged and documented
- ◆ Live demo capability: can showcase agent learning during placement interviews
- ◆ iThynx Reinforcement Learning Certification
- ◆ Capstone: Multi-agent simulation project — original research potential for paper submission

MODULE 04



ARTIFICIAL INTELLIGENCE

The Full Spectrum — Reasoning, Vision, Language, Planning, and Ethics

OVERVIEW

Artificial Intelligence is the broadest and most foundational module — covering everything from classical search algorithms and knowledge systems to modern computer vision and NLP. Students gain the complete mental model of how AI systems are architected, which makes them dramatically better at understanding any new AI tool or paper they encounter throughout their careers. This is the module that develops AI intuition.

Universal

AI principles apply across every specialisation

Computer Vision

Used in manufacturing, healthcare, security, retail

NLP

Every company with customers now needs NLP engineers

Ethics

Fastest-growing AI subfield — and exam boards now test it

CURRICULUM

CONCEPT & SKILL	HANDS-ON PROJECT / LAB
Search & Optimization — A*, Minimax, Alpha-Beta	Build a chess-playing AI that students can play against
Knowledge Representation & Expert Systems	Medical diagnosis expert system — rule-based + probabilistic
Computer Vision — Object Detection (YOLOv8)	Real-time webcam object detection — demo runs in the classroom
Natural Language Processing — NLP Pipeline	Text summarizer + keyword extractor web app on real articles
Speech Recognition & Text-to-Speech	Voice-controlled assistant prototype — speaks back to the user
AI Planning & Scheduling	Autonomous task planner for a logistics / delivery use case
Ethics, Bias & Responsible AI Practice	Bias audit on a public dataset — findings presented as report

TOOLS & TECH



YOLOv8 / OpenCV



spaCy / NLTK



Whisper / TTS



Prolog / CLIPS



WHAT STUDENTS WALK AWAY WITH

- ◆ A fully functional AI application combining vision + language + reasoning components
- ◆ Live object detection demo — extremely impressive in placement interviews and fests
- ◆ iThynx Artificial Intelligence Certification
- ◆ Capstone: AI decision-support tool for a real business domain — presented to jury panel

MODULE 05



GENERATIVE AI + AGENTIC AI

Create, Reason, Act — The Hottest Skill in the Industry Right Now

OVERVIEW

Generative AI creates new content — text, images, code, audio. Agentic AI goes further: it reasons, plans, and takes autonomous actions across tools and APIs to complete complex multi-step goals without human involvement. ChatGPT, Gemini, Copilot, Midjourney — this is the module that teaches students how all of it actually works, and more importantly, how to build with it. This is the single highest-demand skillset in hiring right now.

#1

Most-searched AI skill on LinkedIn in 2024–25

1000%

Growth in GenAI job postings in 18 months

RAG

Architecture behind every enterprise AI chatbot

Agentic AI

Next frontier — students learn this before most engineers have heard of it

CURRICULUM

CONCEPT & SKILL	HANDS-ON PROJECT / LAB
GenAI Foundations — Diffusion Models, GANs, VAEs	Generate custom images using Stable Diffusion on own prompts
Prompt Engineering — Zero/Few-Shot, Chain-of-Thought	Structured prompt library built for 10 real business scenarios
RAG — Retrieval Augmented Generation	Chatbot trained on your own PDF documents — no hallucinations
Fine-Tuning LLMs — LoRA, QLoRA in practice	Fine-tune a small open-source LLM on domain-specific data
LangChain & LlamaIndex Orchestration	Multi-document research assistant — one query, multiple sources
Agentic AI — ReAct Loop, Tool Use, Memory	Agent that browses the web, runs code, sends emails autonomously
Multi-Agent Orchestration — CrewAI / AutoGen	Multi-agent pipeline that completes a real business workflow end-to-end

TOOLS & TECH

 LangChain	 LlamaIndex	 CrewAI / AutoGen	 Stable Diffusion
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WHAT STUDENTS WALK AWAY WITH

- ◆ A production-grade RAG chatbot deployed on the web — shareable public link
- ◆ An autonomous AI agent that completes real-world tasks without human prompting
- ◆ iThynx Generative & Agentic AI Certification
- ◆ Capstone: Multi-agent system that automates a complete business workflow — live demo day

MODULE 06



LARGE LANGUAGE MODELS

Under the Hood of GPT, Gemini & Llama — Build Your Own, Deploy It Live

OVERVIEW

Large Language Models are the most transformative technology of this decade — and almost nobody understands how they actually work. This module goes deep: students understand transformer architecture internally, learn how models are trained and aligned, and build production systems on top of them. This is AI engineering at its most advanced — and the students who complete this module are genuinely rare in the job market.

**GPT-4,
Gemini,
Llama 3**

The models this module demystifies completely

Fine-Tuning

The skill that makes a generic LLM useful for any domain

**RAG +
RLHF**

The two techniques behind every enterprise LLM product

Rare

<1% of engineers understand LLM internals — your students will

CURRICULUM

CONCEPT & SKILL	HANDS-ON PROJECT / LAB
Transformer Architecture — Attention is All You Need	Code a mini-Transformer from scratch in PyTorch — no magic
Tokenization, Embeddings & Context Windows	Visualize how LLMs read and process text at the token level
Prompt Engineering at Scale — Systematic Approach	Build a structured prompt library with measurable eval metrics
Fine-Tuning — Supervised Fine-Tuning (SFT)	Full fine-tuning of a small LLM on a custom instruction dataset
Parameter-Efficient Fine-Tuning — LoRA & QLoRA	Fine-tune Llama 3 on a laptop — efficient and reproducible
RLHF & DPO — Aligning LLMs to be Helpful	Understand exactly how ChatGPT was trained to follow instructions

LLM Evaluation — BLEU, ROUGE, Human Eval	Run a rigorous head-to-head model comparison study with report
LLM Deployment & MLOps — vLLM, Ollama	Serve a fine-tuned model via API — live endpoint, load tested

TOOLS & TECH

 Llama 3 / Mistral	 HF PEFT / TRL	 vLLM / Ollama	 Weights & Biases
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WHAT STUDENTS WALK AWAY WITH

- ◆ A fine-tuned LLM served via API — students own the weights and the endpoint
- ◆ Deep internals knowledge: can explain transformer attention to any interviewer
- ◆ iThynx LLM Engineering Certification
- ◆ Capstone: End-to-end fine-tuned LLM product — dataset curation → training → deployment → demo

MODULE 07



FULL-STACK AI DEVELOPMENT

Build Complete AI Products — From Model to Live Web App in the Real World

OVERVIEW

An AI model trapped in a Jupyter notebook is not a product. Full-Stack AI Development teaches students to wrap their models in beautiful, functional web applications — complete with user authentication, databases, real-time inference, and cloud hosting. Students graduate from this module with apps that are actually live on the internet — not just screenshots. This is the module that turns an AI student into an AI engineer that companies hire on the spot.

Full-Stack + AI

The rarest and highest-paid engineering profile in 2025

Live App

Students demo a real URL — not a PowerPoint — on Demo Day

SaaS-Ready

Students understand the full AI product lifecycle end-to-end

Hire-Ready

Our Full-Stack AI graduates average 40% higher callbacks

CURRICULUM

CONCEPT & SKILL	HANDS-ON PROJECT / LAB
Next.js & Server Components — Hydration & SSR	Build a production-ready application from scratch using Next.js App Router — zero lag
State Management, Hooks & Context API	Implement clean global state architectures and custom hooks to manage complex user workflows
API Architecture & Database Design — REST & GraphQL	Design optimized database schemas and high-throughput APIs for instant data delivery
Authentication & Security — OAuth & JWT	Implement robust secure login flows, session handling, and role-based access control
Performance Optimization — Caching & Edge Compute	Deploy Edge functions and implement advanced caching strategies to minimize global latency

CI/CD & Cloud Deployment — Docker & AWS

Containerize your stack and build automated pipelines to deploy the live application safely

TOOLS & TECH



React / Next.js



FastAPI



Vercel / GCP



Pinecone / PostgreSQL



WHAT STUDENTS WALK AWAY WITH

- ◆ A live, publicly accessible AI SaaS product — students share the link in interviews
- ◆ Complete full-stack GitHub portfolio — frontend, backend, AI, database, deployment
- ◆ iThynx Full-Stack AI Developer Certification
- ◆ Capstone: Real AI product — presented at Demo Day to faculty panel and industry guests

MODULE 08



MODERN FULLSTACK

Next-Gen Web Development — Architecture, Scaling & Production Deployment

OVERVIEW

Modern Fullstack development is the backbone of the digital economy — yet very few developers understand how to build systems that scale seamlessly under heavy production loads. This module goes deep: students master next-generation architecture internally, learn how to optimize databases and state management, and deploy highly resilient web applications live. This is fullstack engineering at its most advanced — and the students who complete this module are genuinely rare in the job market.

Next.js, Node, React

The frameworks and libraries this module demystifies completely

Scalability

The skill that makes a generic application ready for any enterprise volume

CI/CD + Cloud

The two techniques behind every production-grade fullstack product

Rare

<1% of developers understand true system scaling — your students will

CURRICULUM

CONCEPT & SKILL	HANDS-ON PROJECT / LAB
Next.js & Server Components — Hydration & SSR	Build a production-ready application from scratch using Next.js App Router — zero lag
State Management, Hooks & Context API	Implement clean global state architectures and custom hooks to manage complex user workflows
API Architecture & Database Design — REST & GraphQL	Design optimized database schemas and high-throughput APIs for instant data delivery
Authentication & Security — OAuth & JWT	Implement robust secure login flows, session handling, and role-based access control
Performance Optimization — Caching & Edge Compute	Deploy Edge functions and implement advanced caching strategies to minimize global latency

CI/CD & Cloud Deployment — Docker & AWS

Containerize your stack and build automated pipelines to deploy the live application safely

TOOLS & TECH**Next.js / React****Node.js / Express****PostgreSQL / Prisma****Docker / AWS****WHAT STUDENTS WALK AWAY WITH**

- ◆ A production-ready Fullstack application served live — students own the codebase and the live deployment
- ◆ Deep architecture knowledge: can explain hydration, edge caching, and system scaling to any interviewer
- ◆ iThynx Fullstack Engineering Certification
- ◆ Capstone: End-to-end scalable web product — database design → API build → secure auth → live deployment

MODULE 09



MERN STACK DEVELOPMENT

MongoDB, Express, React & Node — Build, Secure, and Deploy Production-Ready Web Apps

OVERVIEW

The MERN stack is the absolute industry standard for building modern JavaScript-driven applications — yet few developers know how to tie MongoDB, Express, React, and Node together securely and efficiently. This module goes deep: students master asynchronous patterns internally, learn how to model relational data in a non-relational database, and build complex single-page apps. This is full-stack JavaScript at its most advanced — and the students who complete this module are genuinely rare in the job market.

React & Node.js

The core runtime and frontend library this module demystifies completely

Schema Design

The skill that makes flexible MongoDB collections useful for any domain

RESTful APIs

The communication architecture behind every enterprise MERN product

Rare

<1% of developers understand advanced MERN authentication & optimization — your students will.

CURRICULUM

CONCEPT & SKILL	HANDS-ON PROJECT / LAB
Node.js Internals & Express.js Routing	Build a high-performance backend server from scratch — master middleware and error handling
MongoDB Schema Design & Mongoose ODM	Model complex real-world data relationships using validation, aggregation, and indexing
React Single Page Applications (SPAs)	Build dynamic, responsive user interfaces with optimized state management and routing
JWT Authentication & Middleware Security	Implement robust secure login flows, password hashing, and protected API routes
State Management with Redux Toolkit	Handle global application state and asynchronous data fetching across large interfaces

Testing & Deployment — Jest & Heroku/Render

Write automated unit tests for your API and deploy the unified application to production live

TOOLS & TECH**MongoDB / Mongoose****Express.js****React / Redux****Node.js****WHAT STUDENTS WALK AWAY WITH**

- ◆ A complete MERN Stack application served live — students own the database, server, and client deployment
- ◆ Deep architecture knowledge: can explain asynchronous event loops, component life cycles, and database modeling to any interviewer
- ◆ iThynx MERN Stack Developer Certification
- ◆ Capstone: End-to-end fullstack JavaScript product — database setup → custom API routing → state management → live deployment

MODULE 10



DATA ENGINEERING & DATA SCIENCE

Infrastructure at Scale — Architecting Robust Batch and Real-Time Data Ecosystems & Predictive Analytics

OVERVIEW

Raw data is useless without the infrastructure to move, clean, and store it—and the analytical models to extract value from it. This module dives deep into data architecture and data science: students design distributed systems, build production-grade ETL pipelines, develop predictive machine learning models, and master stream processing. This is data engineering and data science at its absolute limit — transforming chaotic data streams into corporate goldmines and predictive intelligence.

Kafka, Spark, Airflow & Scikit-Learn

The exact stack driving modern data-driven enterprises.

ETL pipeline Predictive

The skill that makes flexible MongoDB collections useful for any domain

Streaming Batch, Statistical inference

The two processing paradigms behind real-time financial and event analytics.

Architect

<1% of developers can design fault-tolerant data lakes — your students will.

CURRICULUM

CONCEPT & SKILL	HANDS-ON PROJECT / LAB
Distributed Systems — HDFS & MapReduce Internals	Configure a multi-node cluster to process terabytes of data across distributed environments
Data Warehousing — Schema Design & Indexing	Design optimized Star and Snowflake schemas for complex corporate reporting requirements
ETL Pipelines — Workflow Orchestration with Airflow	Build a resilient, automated data pipeline that extracts, transforms, and loads API data
Big Data Compute — Optimization with Apache Spark	Code a distributed data cleaning engine that handles data skew and memory caching

Stream Processing — Real-Time Event Driven Data	Build a continuous real-time data ingestion pipeline using Apache Kafka and Spark Streaming
Data Lakes — Delta Lake & Modern Storage Layers	Implement ACID transactions and time-travel querying on top of unstructured cloud storage

TOOLS & TECH

 Apache Spark , Scikit	 Apache Airflow , Numpy	 Apache Kafka , BI	 Snowflake
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WHAT STUDENTS WALK AWAY WITH

- ◆ A complete MERN Stack application served live — students own the database, server, and client deployment
- ◆ Deep architecture knowledge: can explain asynchronous event loops, component life cycles, and database modeling to any interviewer
- ◆ iThynx MERN Stack Developer Certification
- ◆ Capstone: End-to-end fullstack JavaScript product — database setup → custom API routing → state management → live deployment

MODULE 11



JAVA PROGRAMMING & ENTERPRISE ARCHITECTURE

Under the Hood of the JVM, Spring Boot & Microservices — Build Scalable Systems, Deploy to Cloud

OVERVIEW

Java powers the backbone of global enterprise infrastructure, yet few developers truly understand its runtime mechanics. This module goes deep: students master JVM internals and memory management, build production-grade backends using Spring Boot, and architect resilient microservices. This is backend engineering at scale—and the students who complete this module possess the highly optimization-focused skillset required by elite tech teams.

JVM, JDK & JRE The runtime environment this module demystifies completely.	Multi-threading The core skill required to build high-concurrency, non-blocking applications.	Spring Boot + K8s The golden standard framework pair behind enterprise cloud ecosystems.	Top 5% Fewer than 5% of engineers write optimized, garbage-collector-aware Java—your students will.
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CURRICULUM

CONCEPT & SKILL	HANDS-ON PROJECT / LAB
JVM Architecture — Memory Management & Garbage Collection	Profile a live Java application, track heap dumps, and tune GC parameters to fix a simulated memory leak.
Object-Oriented Design & Design Patterns	Refactor a legacy monolithic code snippet into clean, decoupled code using Factory, Singleton, and Strategy patterns.
Concurrency & Multithreading — The Concurrency API	Build a high-throughput, multi-threaded web scraper using Executors, ThreadPools, and CompletableFuture.

Spring Boot Foundations — Dependency Injection & Data JPA	Develop a secure, RESTful e-commerce backend API connecting to a PostgreSQL database via Spring Data JPA.
Enterprise Microservices — Service Discovery & Gateway	Architect a distributed microservices system featuring a Netflix Eureka Discovery Server, Spring Cloud Gateway, and Resilience4j circuit breakers.
Unit Testing & CI/CD — Mockito & Dockerization	Write robust unit tests achieving 90%+ code coverage with JUnit 5 and Mockito, then containerize the app via Docker.

TOOLS & TECH

 Java 21 / OpenJDK	 Spring Boot 3 / Hibernate	 Docker / Kubernetes	 JUnit 5 / Mockito
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WHAT STUDENTS WALK AWAY WITH

- ◆ A production-ready, microservices-based backend served via cloud API—students own the source code and deployment pipeline.
- ◆ Deep internals knowledge: can explain Java memory layouts (Heap vs. Stack) and thread safety synchronization to any interviewer.
- ◆ iThynx Advanced Java Developer Certification.
- ◆ Capstone: End-to-end enterprise Java application—from object-oriented design and multi-threading implementation to database caching, cloud deployment, and performance monitoring dashboard setup.



MODULE 12

DATA STRUCTURES & ALGORITHMS IN JAVA

Cracking the Coding Interview — Master Problem-Solving Patterns, Optimize Complexity, Ace MAANG Assessments

OVERVIEW

Writing code that works is easy; writing code that scales efficiently under massive enterprise loads is what defines an elite engineer. This intensive module takes students deep into core data structures, advanced algorithmic strategies, and algorithmic analysis using Java. By translating raw logic into highly optimized, edge-case-proof code, students master the exact technical problem-solving frameworks demanded by top-tier product companies.

LeetCode 150+

The core industry-standard interview problems demystified completely.

Time & Space

The skill that makes a programmer capable of optimizing runtime and memory allocation.

Java Collections API

Under the hood of built-in data structures like HashMap, PriorityQueue, and ArrayDeque.

Rare

<1% of candidates can accurately trace complex Dynamic Programming or Graph states—your students will.

CURRICULUM

CONCEPT & SKILL	HANDS-ON PROJECT / LAB
Asymptotic Analysis & Linear Structures (Arrays & Strings)	Implement multi-pointer and sliding window algorithms to optimize brute-force processing down to $O(N)$ runtime.
Custom Memory Linking — Singly, Doubly, & Circular Linked Lists	Code a memory-efficient custom Linked List from scratch and implement complex pointer-manipulation algorithms like cycle detection.
Restricted Access Structures — Stacks & Queues	Build a low-latency expression evaluation engine and design a custom "Min Stack" structure operating in constant $O(1)$ time.

Hierarchical Structures & Hierarchical Traversal — Binary Trees & BSTs	Develop a recursive evaluation engine to handle Tree traversals (Inorder, Preorder, Postorder) and construct self-balancing search logic.
Advanced Data Structures — Heaps, HashMaps, & Tries	Build an autocomplete search engine using a Trie data structure, and use Java's PriorityQueue to solve stream-median challenges.
Non-Linear Graphs & Complex Optimization — Dynamic Programming & Graph Algos	Implement BFS, DFS, and Dijkstra's algorithm for pathfinding, then design highly optimized matrix calculators using bottom-up DP.

TOOLS & TECH

 Java 21 Collections	 IntelliJ Profiler	 JUnit 5 Benchmarks	 LeetCode / HackerRank Env
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WHAT STUDENTS WALK AWAY WITH

- ◆ A self-built, fully optimized repository containing benchmarked solutions to over 150+ production-level DSA problems.
- ◆ Deep complexity knowledge: can break down, defend, and mathematically prove the Big-O time and space trade-offs to any technical interviewer.
- ◆ iThynx Technical Problem Solving & DSA Certification.
- ◆ Capstone: A fully functional technical mock interview simulation engine—featuring custom memory profiling, code execution run-times, dynamic testing datasets, and automated optimal space complexity scoring.

MODULE 12



INTERPERSONAL COMMUNICATION & INTERVIEW COACHING

Mastering the Art of 1-on-1 Mentorship, Boosting Confidence, and Conducting High-Impact Mock Interviews

OVERVIEW

Interpersonal communication is the most critical yet overlooked professional skill—and true mentors who can build confidence in others are incredibly rare. This module goes deep: students master the psychology of constructive feedback, learn how to run transformative 1-on-1 mock interviews, and discover how to systematically dismantle a candidate's interview anxiety. This is communication engineering at its most impactful—and the individuals who complete this module become exceptionally powerful leaders and mentors in the market.

Active Listening & Empathy The core foundation this module demystifies completely to read non-verbal cues.	Constructive Feedback The skill that turns a bad mock interview into a massive growth opportunity.	STAR Method The two powerful techniques behind building psychological safety and structural clarity..	Elite Mentors <1% of professionals know how to coach confidence—your students will.
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CURRICULUM

CONCEPT & SKILL	HANDS-ON PROJECT / LAB
The Psychology of Confidence — Building Psychological Safety	Roleplay a high-anxiety coaching icebreaker from scratch — no awkwardness
Active Listening, Non-Verbal Cues & Empathy Windows	Analyze and tag interview video recordings to spot hidden communication blocks
Constructive Feedback at Scale — The Feedback Sandwich Approach	Build a structured feedback evaluation rubric with measurable growth metrics

Mock Interview Framing — Behavioral & Technical Alignment	Conduct a full 1-on-1 live mock interview using custom target job descriptions
Anxiety De-escalation — Overcoming the "Brain Freeze"	Coach a panicking candidate through live roadblocks — efficient and reproducible
STAR Method & Delivery — Aligning Answers to Be Punchy	Understand exactly how elite candidates are trained to tell compelling stories



WHAT STUDENTS WALK AWAY WITH

- ◆ A personalized, field-tested mock interview evaluation rubric — students own their coaching framework..
- ◆ Deep interpersonal knowledge: can confidently diagnose and fix communication flaws in any interviewee
- ◆ Professional Interview Coach Certification
- ◆ Capstone: End-to-end transformation case study — initial assessment , anxiety mapping , live mock drills , job-ready placement demo

MODULE 13



AWS CLOUD COMPUTING

The Complete Cloud Infrastructure — Compute, Networking, Security, DevOps & Containers

OVERVIEW

AWS Cloud Computing is a comprehensive module designed to equip students with industry-ready cloud engineering skills. Students will learn how to build, deploy, secure, monitor, and automate cloud infrastructure using Amazon Web Services. The curriculum covers compute, networking, storage, security, DevOps, serverless computing, databases, monitoring, and container orchestration with extensive hands-on labs and real-world projects.

Cloud Infrastructure Build scalable environments	DevOps Automation CI/CD deployment pipelines	Containers Docker , ECR, Kubernetes	Industry Ready Know how to manage cloud platform for an industry
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CURRICULUM

CONCEPT & SKILL	HANDS-ON PROJECT / LAB
Introduction to AWS	AWS Account Setup, Global Infrastructure, Regions & Availability Zones
IAM (Identity & Access Management)	Create Users, Groups, Roles, Policies & MFA
EC2 (Elastic Compute Cloud)	Launch Linux Instances, SSH using PuTTY, Host Web Applications
AWS Networking (VPC)	Design Custom VPC, Public & Private Subnets, Internet Gateway, NAT Gateway
AWS Security	Security Groups, NACLs, IAM Best Practices
Route 53	Domain Registration, DNS Routing, Hosted Zones
Amazon S3	Static Website Hosting, Versioning, Lifecycle Policies

CloudFormation	Infrastructure as Code (IaC) Templates
Elastic Beanstalk	Deploy Full-Stack Applications Automatically
CodeCommit	Git Repository Management on AWS
CodePipeline (CI/CD)	Automated Build-Test-Deploy Pipeline
CodeBuild	Build & Test Applications Automatically
CodeDeploy	Zero-Downtime Application Deployment
CloudWatch	Metrics, Dashboards, Monitoring & Alarms
AWS Lambda	Build Serverless Applications
EventBridge (CloudWatch Events)	Event-Driven Automation
CloudTrail	Audit AWS Account Activities
DynamoDB	Build NoSQL Applications
Elastic Container Service (ECS)	Deploy Docker Containers
Elastic Container Registry (ECR)	Store & Manage Docker Images
Elastic Kubernetes Service (EKS)	Kubernetes Cluster Deployment
CloudWatch Logs	Centralized Log Management
Secrets Manager	Secure Passwords & API Keys
Auto Scaling	High Availability & Automatic Scaling
Relational Database Service (RDS)	MySQL/PostgreSQL Database Deployment
Elastic Load Balancer (ELB)	High Availability & Traffic Distribution
AWS Systems Manager	Remote Server Management & Automation
Cloud Migration Strategies	Lift & Shift, Replatform, Refactor Migration Labs

 AWS Console	 S3 Buckets	 Docker + ECS + ECR	 Kubernetes (EKS)
 Lambda	 EC2	 VPC	 Secrets Manager



WHAT STUDENTS WALK AWAY WITH

- ❖ Industry-ready AWS Cloud Engineer skills
- ❖ Hands-on experience deploying scalable cloud applications
- ❖ Complete CI/CD pipeline implementation using AWS DevOps tools
- ❖ Serverless application development using AWS Lambda
- ❖ Docker & Kubernetes deployment experience (ECS & EKS)
- ❖ Infrastructure as Code (CloudFormation)
- ❖ Cloud monitoring, logging & security best practices
- ❖ AWS networking architecture design (VPC, ELB, Auto Scaling)
- ❖ Capstone Project: Deploy a highly available, scalable, secure cloud application using AWS services
- ❖ AWS Cloud Computing Certification from the institute

08 OUR PARTNERSHIP PROMISE TO YOUR INSTITUTION

✓ Zero Disruption	We work around your academic timetable. Our trainers come to you. You manage nothing.
✓ Zero Faculty Burden	All content, labs, grading, and session management handled entirely by our team.
✓ Full Documentation	MOU, session logs, attendance, outcome reports — everything needed for NBA/NAAC files.
✓ Measurable Outcomes	We commit to specific, documented student outcomes. Not vague promises.
✓ Pilot First	We propose starting with a small pilot cohort — no long-term commitment required upfront.
✓ Transparent Pricing	All costs disclosed upfront. No hidden charges. Flexible models to fit your budget.

09 PROPOSED NEXT STEPS

STEP 1	This meeting — understand your institution's needs, batch size, and academic calendar
STEP 2	We share a customised curriculum map aligned to your department and student profile
STEP 3	Pilot cohort — small batch, one module — let results speak before full commitment
STEP 4	Review outcomes, gather student and faculty feedback
STEP 5	MOU signing and full program rollout — on your terms, your pace

Let's Build Something Remarkable Together

Your students deserve to graduate ready for the world as it is — not as it was.

We're not asking for a long-term commitment today.

Just one pilot cohort. Let the results make the case.

✉ contact www.ithynx.com · ☎ +91 63699 83585 · 🌐 www.ithynx.com

** All program content, curriculum, tools, batch sizes, and investment details are fully customisable to your institution's specific requirements. This proposal is prepared exclusively for your institution and is not shared with any competing college in your geography.*